

EXERCISE SHEET 8

GROUP AND RING THEORY

MAIN QUESTIONS - TO BE DONE BEFORE THE SEMINAR

8.1. Let M be a Noetherian and Artinian R -module, for a ring R . Show that the following are equivalent for an R -homomorphism $f : M \rightarrow M$:

- (i) f is bijective;
- (ii) f is injective;
- (iii) f is surjective.

8.2. Let R be a Noetherian R -module. Show that the ring of $n \times n$ matrices $M_n(R)$ over R is also Noetherian as an $M_n(R)$ -module.

EXTRA QUESTIONS

8.3. Let R be a Noetherian ring with unity and let M be a finitely generated R -module. Show that all submodules of M are finitely generated.

8.4. Show that if R_i , for $i \in \mathbb{N}$, is an infinite family of non-zero rings and if R is the direct sum of this family (R_i) , for $i \in \mathbb{N}$, then R cannot be Noetherian.